

CTA200 Assignment 3 Results

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1 Question 1: Mandelbrot Iteration

For Question 1, I investigated the iteration

$$z_{i+1} = z_i^2 + c,$$

with initial value $z_0 = 0$. The complex number c was constructed from points (x, y) , where x was used as the real component and y was used as the imaginary component. For each point, I iterated the equation up to 1000 times and defined divergence as the condition $|z| > 2$.

Figure 1 shows the points in the complex plane that diverged under this iteration. Figure 2 shows the number of iterations required for divergence, using a logarithmic colour scale.

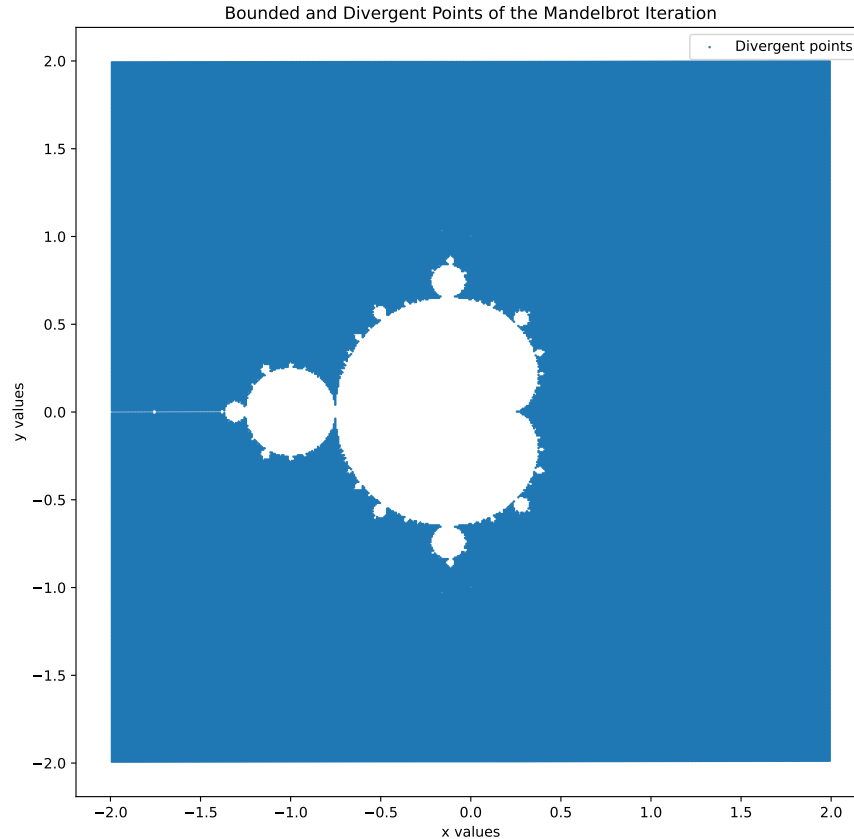


Figure 1: Points in the complex plane that diverged under the Mandelbrot iteration.

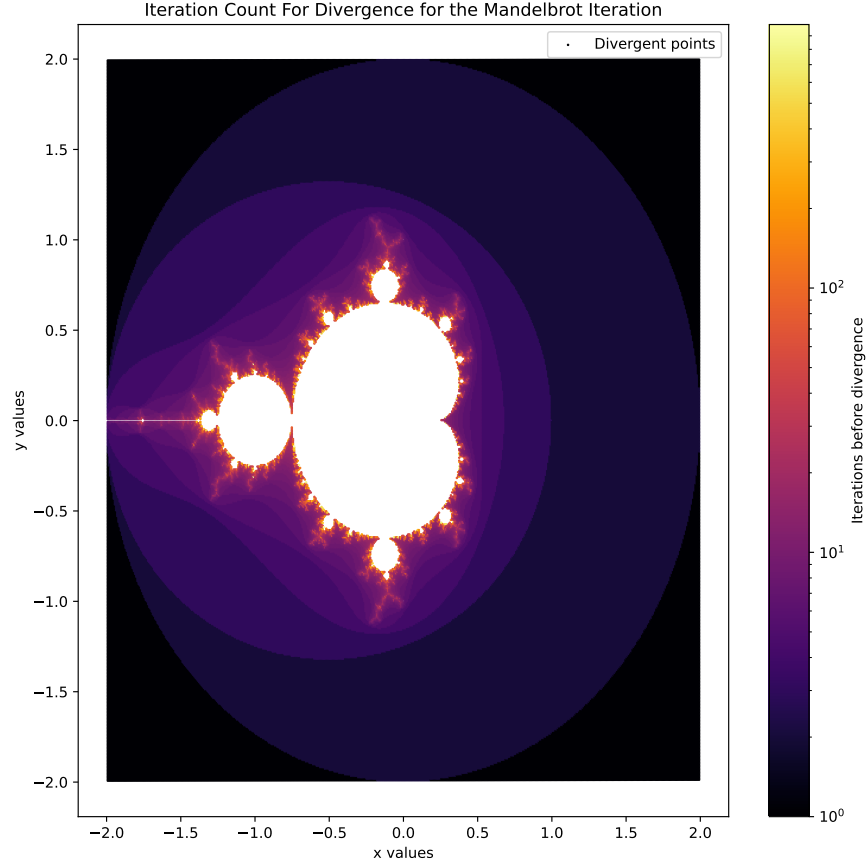


Figure 2: Iteration count required for divergence in the Mandelbrot iteration.

2 Question 2: Lorenz System

For Question 2, I solved the Lorenz system

$$\begin{aligned}\frac{dX}{dt} &= -\sigma(X - Y), \\ \frac{dY}{dt} &= rX - Y - XZ, \\ \frac{dZ}{dt} &= -bZ + XY.\end{aligned}$$

I used the parameter values

$$\sigma = 10, \quad r = 28, \quad b = \frac{8}{3},$$

with Lorenz's initial condition

$$W_0 = [0, 1, 0].$$

The system was integrated numerically using `solve_ivp` from $t = 0$ to $t = 60$. Dense output was used so that the solution could be evaluated at closely spaced time values for plotting.

Figure 3 reproduces Lorenz's Figure 1 by plotting Y against the iteration number

$$N = \frac{t}{\Delta t},$$

where $\Delta t = 0.01$. The three panels show the intervals $N = 0$ to 1000, $N = 1000$ to 2000, and $N = 2000$ to 3000.

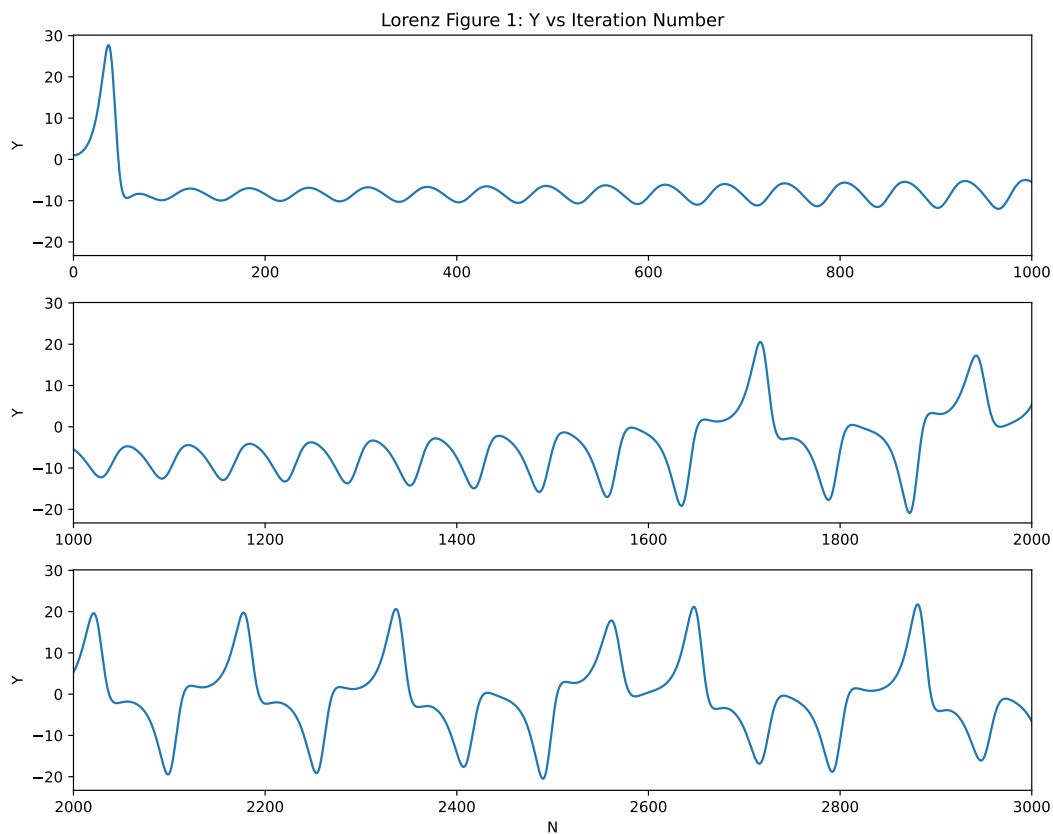


Figure 3: Reproduction of Lorenz's Figure 1, showing Y as a function of iteration number N .

Figure 4 reproduces Lorenz's Figure 2. The solution was evaluated for $14 \leq t \leq 19$, corresponding to $1400 \leq N \leq 1900$. The top panel shows the projection onto the Y - Z plane, while the bottom panel shows the projection onto the Y - X plane.

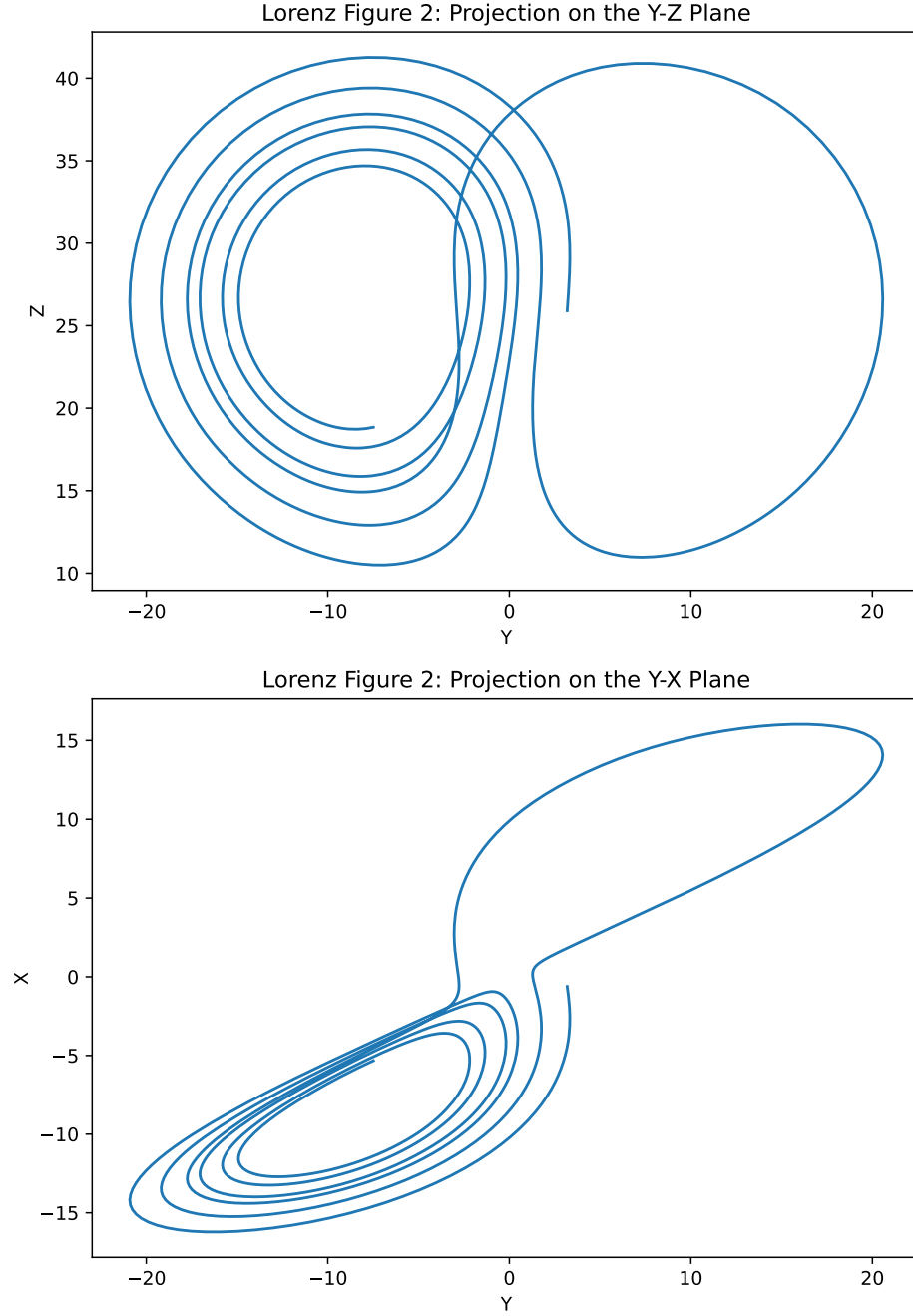


Figure 4: Reproduction of Lorenz's Figure 2, showing projections of the Lorenz trajectory.

To investigate the sensitivity of the system to initial conditions, I solved the Lorenz system again using the slightly perturbed initial condition

$$W'_0 = [0, 1.00000001, 0].$$

I then calculated the distance between the original and perturbed solutions:

$$d(t) = \sqrt{(X' - X)^2 + (Y' - Y)^2 + (Z' - Z)^2}.$$

The result is shown in Figure 5.

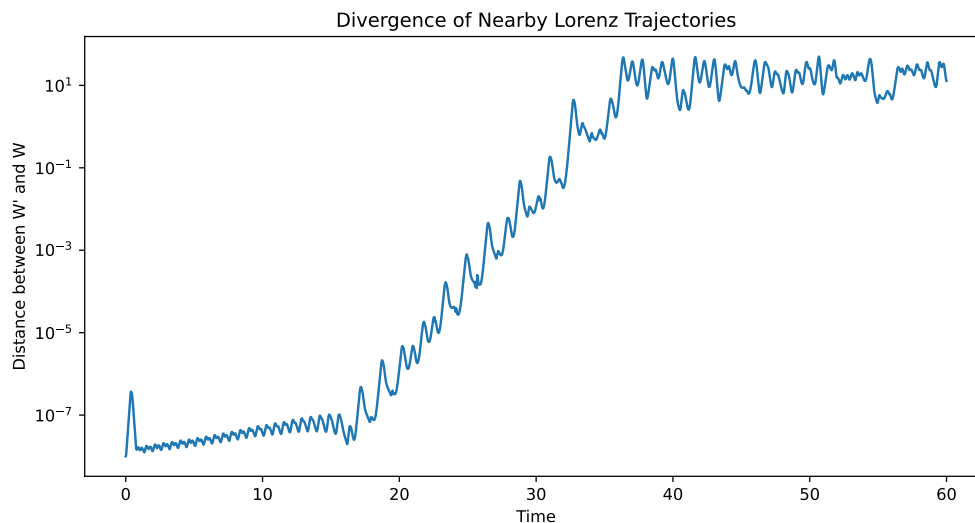


Figure 5: Distance between the original and perturbed Lorenz trajectories as a function of time.

3 Conclusion

The Mandelbrot iteration showed that small changes in the complex parameter c can strongly affect whether the sequence remains bounded or diverges. Similarly, the Lorenz system showed sensitive dependence on initial conditions. Two trajectories starting only 10^{-8} apart eventually separated significantly, which show just how even a little change in the initial conditions of a system can lead to very different evolutions of the system.